Ciencia y Tecnología
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THE ARCHITECTURE OF UNCERTAINTY

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WHAT IS CONDITIONAL PROBABILITY?

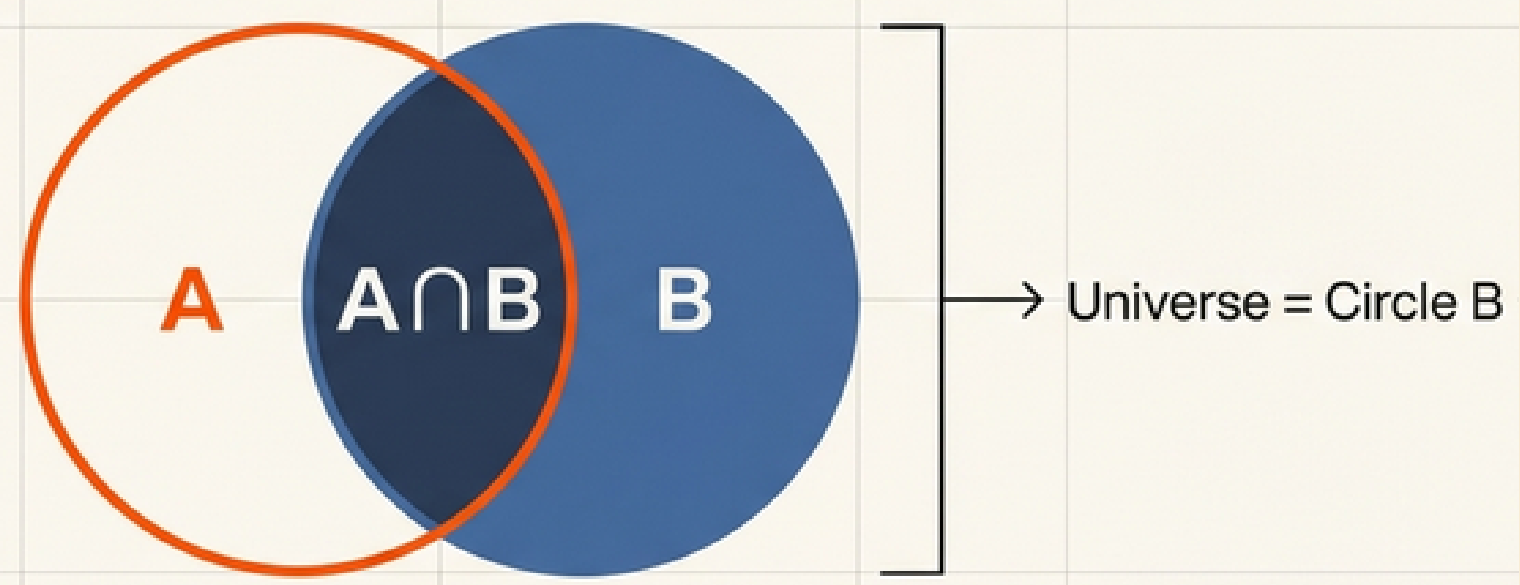
It is the updated probability of an event A after we learn that another event B has occurred.

A major use of probability in statistical inference is updating probabilities when specific events are observed.

Notation: Pr(A|B). Read as "the conditional probability of A given B"

THE DEFINITION OF CONDITIONAL PROBABILITY

Among a large number of repetitions where event B occurs, the proportion in which A also occurs is approximately Pr(A|B).



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

FROM CONDITIONAL TO INTERSECTION

To find the probability that two events occur simultaneously (or sequentially).

$$Pr(A \cap B) = Pr(B)\ Pr(A|B)$$

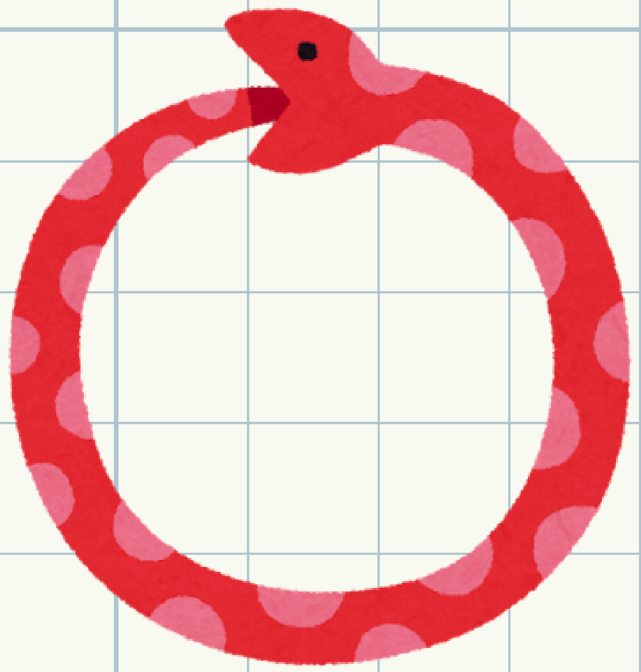
Useful for experiments occurring in stages.

PARTITIONING THE SAMPLE SPACE

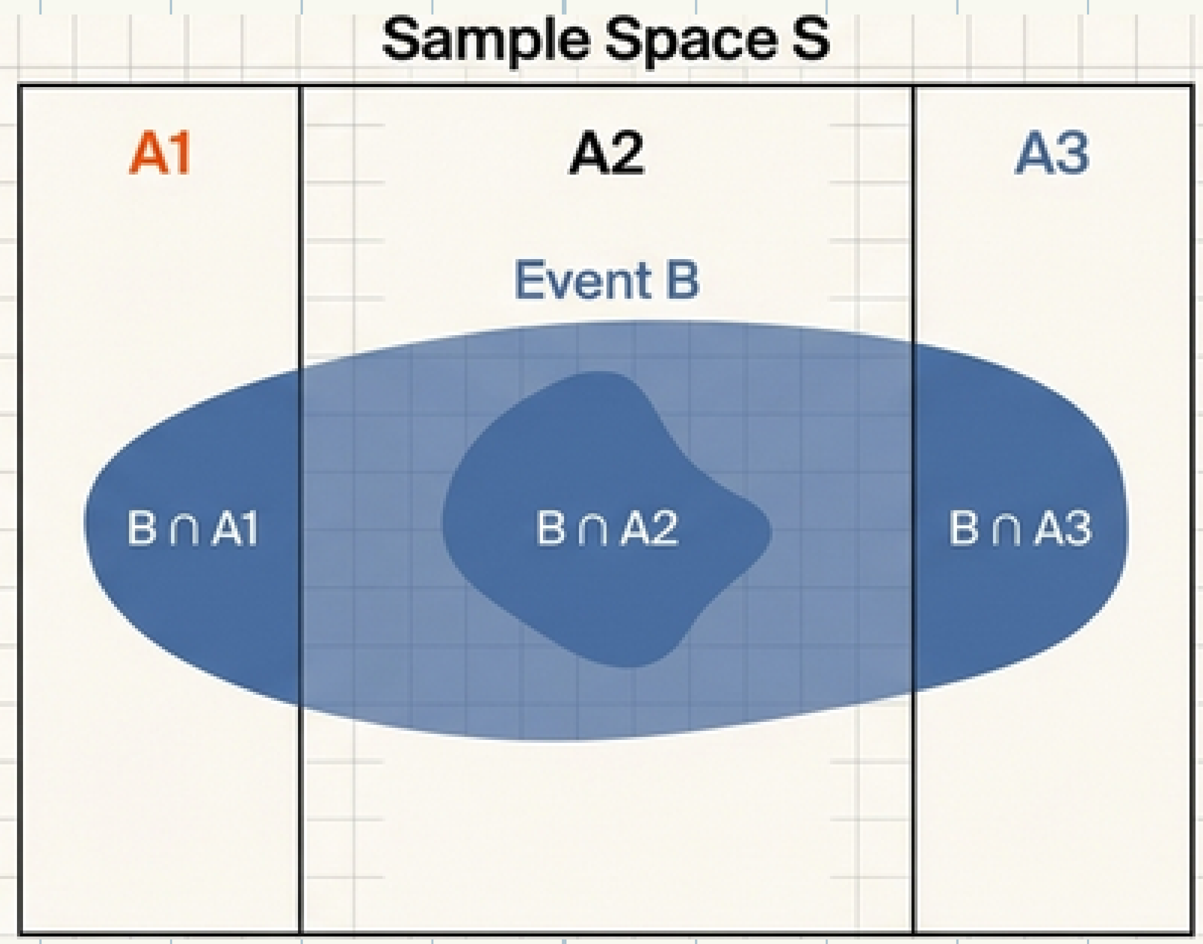
A collection of events B1, . . . , Bk forms a partition of the sample space S if:

The events are disjoint (mutually exclusive).

Their union is the entire sample space



Partitions are chosen to reduce uncertainty. If we learn which Bi occurred, the problem becomes simpler.



THE LAW OF TOTAL PROBABILITY

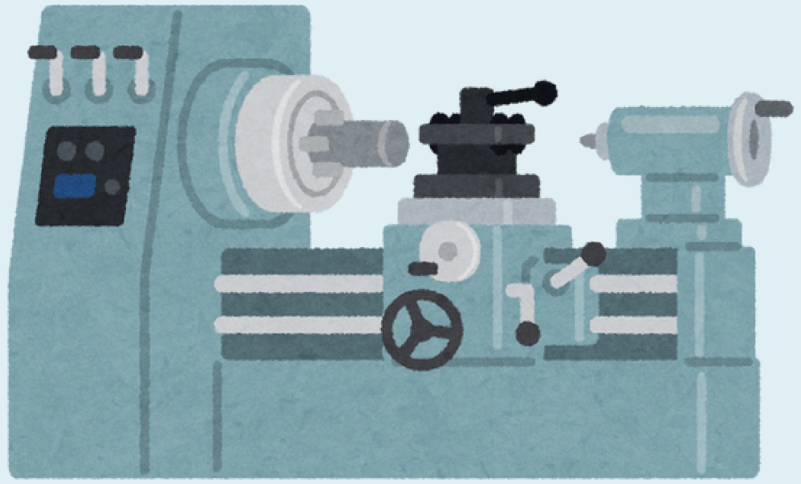
To calculate Pr(A) by breaking it down into parts based on a partition.

$$P(B) = \sum P(B|A_i)P(A_i)$$

To find the total probability of B, we must account for every disjoint path that leads to B.

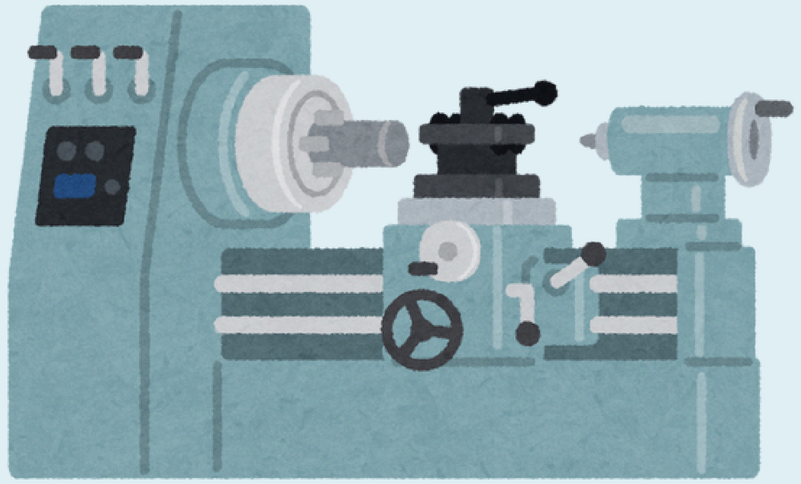
EXAMPLE: MANUFACTURING DEFECTS

Items are produced by different machines .



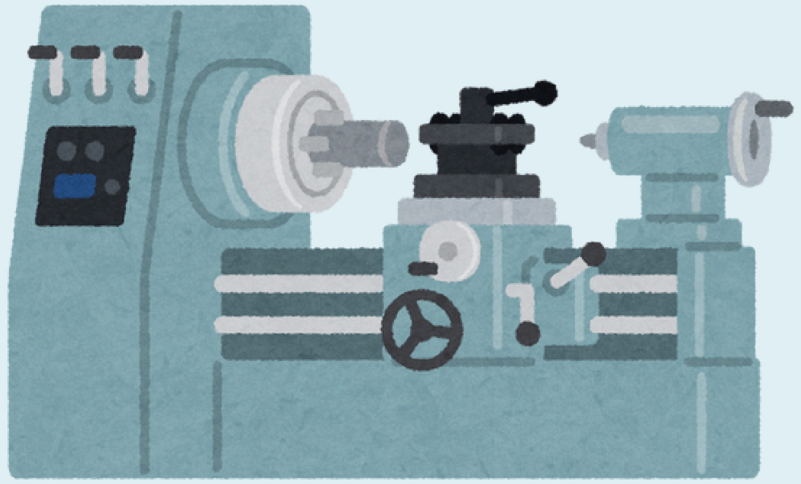
Produces 20% of items.

Defect rate: 1%.



Produces 30% of items.

Defect rate: 2%.



Produces 50% of items.

Defect rate: 3%.

Find the probability a randomly selected item is defective

$$Pr(A) = (0.2)(0.01) + (0.3)(0.02) + (0.5)(0.03) = 0.023$$

The total probability of a defect is 2.3%.

BAYES` THEOREM

To find the probability of a "cause" (Bi) given an observed "effect" (A). We want to reverse the conditional probability.

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B)}$$

BAYESIAN INFERENCE TERMS

Prior Probability (Pr(Bi)): The probability of an event before observing new data (e.g., the probability a machine produced an item before we inspect it).

Posterior Probability (Pr(Bi | A)): The updated probability after observing event A (e.g., the probability the item came from a specific machine after seeing it is defective).

SUMMARY OF KEY CONCEPTS

1.- Conditional Probability: Updates likelihood based on new information (A given B).

2.-Law of Total Probability: Combines conditional probabilities from a partition to find the total probability of an event.

3.- Bayes' Theorem: Uses the Law of Total Probability to reverse conditional probabilities, moving from Prior to Posterior.

LITERATURE

1. Ross, S. M. (2009). Introduction to Probability Models. Academic Press

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3. Devore, J. L. (2010). Probabilidad y estadística para ingeniería y ciencias. 7e.

THANKS!!!